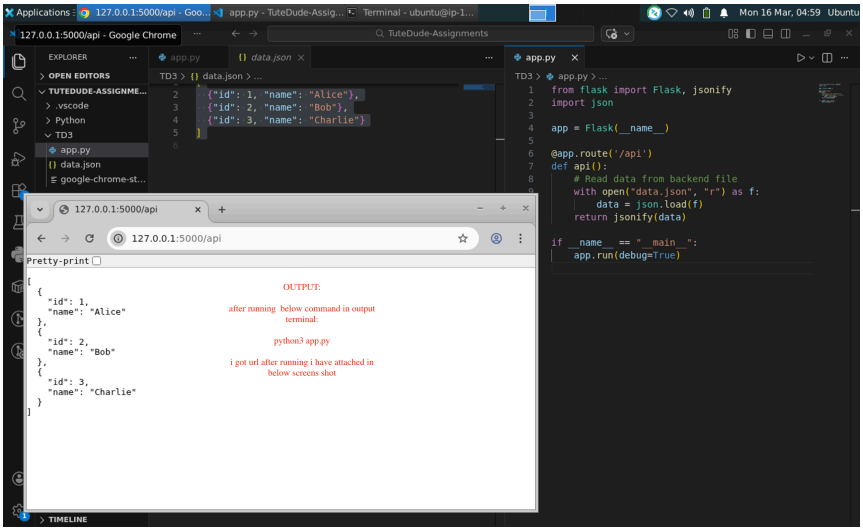
1. **Create a Flask application with an `/api` route. When this route is accessed, it should return a JSON list. The data should be stored in a backend file, read from it, and sent as a response.**

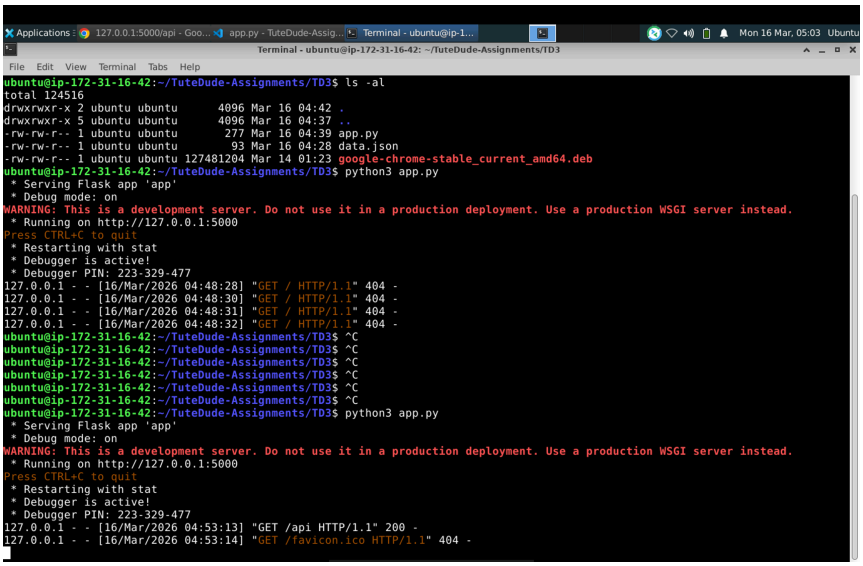
   CODE :

```
from flask import Flask, jsonify

import json

app = Flask(__name__)

@app.route('/api')

def api():

# Read data from backend file

with open("data.json", "r") as f:

        data = json.load(f)

return jsonify(data)

if __name__ == "__main__":

        app.run(debug=True)
```

data.json file :

```
[

  {"id": 1, "name": "Alice"},

  {"id": 2, "name": "Bob"},

  {"id": 3, "name": "Charlie"}

]
```


127.0.0.1:5000/api - Google Chrome
app.py
data.json
{"id": 1, "name": "Alice"},
{"id": 2, "name": "Bob"},
{"id": 3, "name": "Charlie"}
]
from flask import Flask, jsonify
import json
app = Flask(__name__)
@app.route('/api')
def api():
# Read data from backend file
with open("data.json", "r") as f:
data = json.load(f)
return jsonify(data)
if __name__ == "__main__":
app.run(debug=True)
127.0.0.1:5000/api
Pretty-print
[
{
"id": 1,
"name": "Alice"
},
{
"id": 2,
"name": "Bob"
},
{
"id": 3,
"name": "Charlie"
}
]
OUTPUT:
after running below command in output terminal:
python3 app.py
i got url after running i have attached in below screens shot


Terminal - ubuntu@ip-172-31-16-42: ~/TuteDude-Assignments/TD3
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ls -al
total 124516
drwxrwxr-x 2 ubuntu ubuntu 4096 Mar 16 04:42 .
drwxrwxr-x 5 ubuntu ubuntu 4096 Mar 16 04:37 ..
-rw-rw-r-- 1 ubuntu ubuntu 277 Mar 16 04:39 app.py
-rw-rw-r-- 1 ubuntu ubuntu 93 Mar 16 04:28 data.json
-rw-rw-r-- 1 ubuntu ubuntu 127481204 Mar 14 01:23 google-chrome-stable_current_amd64.deb
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ python3 app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 223-329-477
127.0.0.1 - - [16/Mar/2026 04:48:28] "GET / HTTP/1.1" 404 -
127.0.0.1 - - [16/Mar/2026 04:48:30] "GET / HTTP/1.1" 404 -
127.0.0.1 - - [16/Mar/2026 04:48:31] "GET / HTTP/1.1" 404 -
127.0.0.1 - - [16/Mar/2026 04:48:32] "GET / HTTP/1.1" 404 -
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ ^C
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/TD3$ python3 app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 223-329-477
127.0.0.1 - - [16/Mar/2026 04:53:13] "GET /api HTTP/1.1" 200 -
127.0.0.1 - - [16/Mar/2026 04:53:14] "GET /favicon.ico HTTP/1.1" 404 -

OUTPUT:

2. Create a form on the frontend that, when submitted, inserts data into MongoDB Atlas. Upon successful submission, the user should be redirected to another page displaying the message "Data submitted successfully". If there's an error during submission, display the error on the same page without redirection.

CODE:

form.html CODE:

```
<!DOCTYPE html>

<html>

<head>

    <title>Submit Data</title>

</head>

<body>

    <h1>Submit Data</h1>

    <form method="POST">

        <label>Name:</label>

        <input type="text" name="name" required><br><br>

        <label>Email:</label>

        <input type="email" name="email" required><br><br>

        <button type="submit">Submit</button>

    </form>


    {% if error %}
```

```
        <p style="color:red;">Error: {{ error }}</p>
    {% endif %}
</body>
</html>
```

Success.html CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Success</title>
</head>
<body>
    <h1>Data submitted successfully</h1>
</body>
</html>
```

app.py CODE:

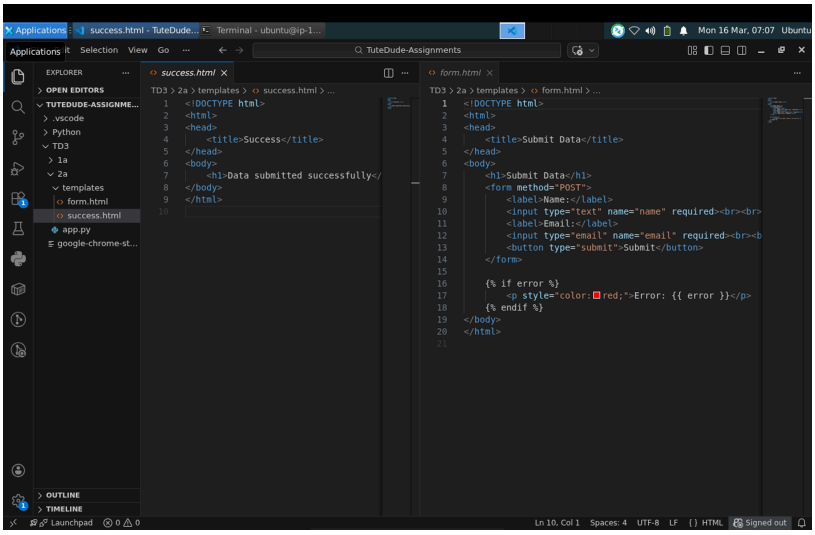
```
from flask import Flask, request, redirect, render_template
from pymongo import MongoClient
```

```
app = Flask(__name__)


# Use your Atlas connection string with your credentials

client = MongoClient(

    "mongodb+srv://
potthuruvenkataumeshchandra_db_user:Q1rNRNmg5mvmoxvl@umeshcluster.zmltzwr.m
ongodb.net/mydatabase?retryWrites=true&w=majority"

)


# Select database and collection

db = client["mydatabase"]

collection = db["mycollection"]


@app.route("/", methods=["GET", "POST"])

def index():

    error = None

    if request.method == "POST":

        try:

            name = request.form.get("name")

            email = request.form.get("email")


            # Insert into MongoDB

            collection.insert_one({"name": name, "email": email})
```

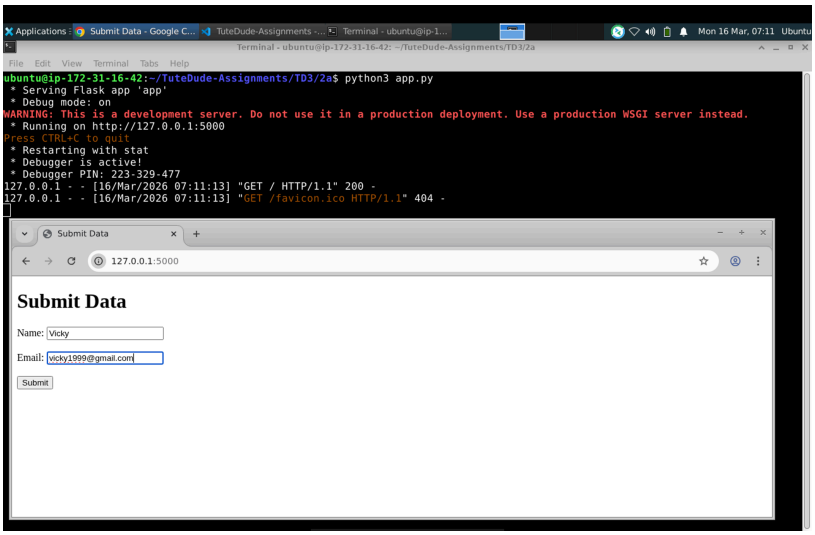
```
            # Redirect to success page
            return redirect("/success")
        except Exception as e:
            error = str(e)

    # Render form with error if any
    return render_template("form.html", error=error)

@app.route("/success")
def success():
    return render_template("success.html")

if __name__ == "__main__":
    app.run(debug=True)
```

OUTPUT:

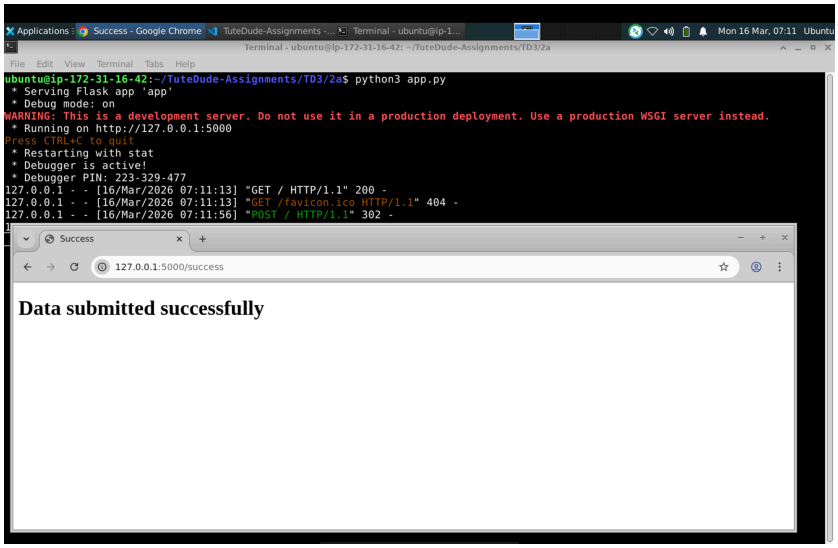
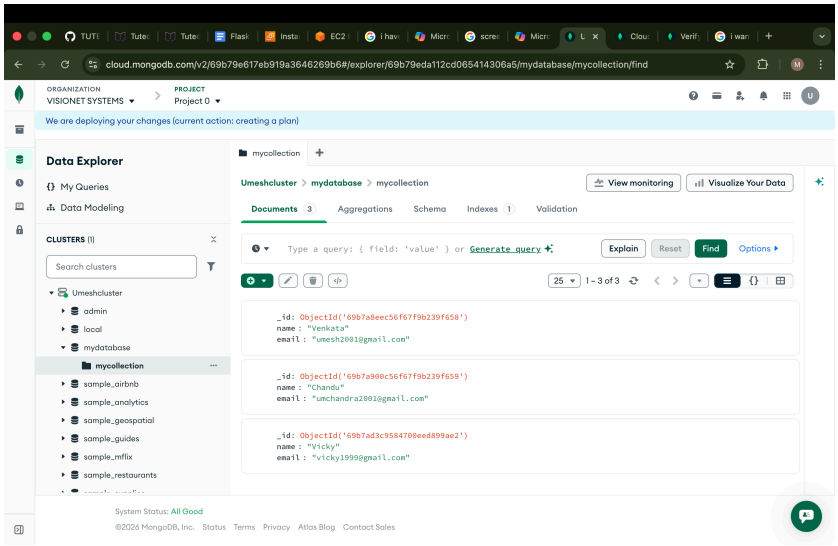
CODE :



APPLICATION OUTPUT:



AFTER ENTERING THE DATA:



MONGO DB DATA: